

Plagiarism Detection (plagiarism)

Giorgio is pursuing a career as a singer-songwriter! However, the music world is tough, and you always need to watch out for plagiarism. In fact, the last song that Alessandro wrote sounds suspiciously similar to one of Giorgio's songs, and now Giorgio wants to prove Alessandro's mischief.



Figure 1: Giorgio writing a new song.

Both Giorgio's and Alessandro's songs are N notes long. Each note¹ is encoded by two integers between 0 and V : its duration d_i , followed by its pitch p_i . Each song is then encoded as a sequence of $2N$ integers. Of course, the plagiarism case will be uncontroversial if the two songs are identical. If not, the plagiarism case will be stronger if Giorgio can find a **short** contiguous sequence of notes that needs to be modified to transform Giorgio's song into Alessandro's.

For such a sequence, if possible, it would be better for Giorgio if he can prove that applying **some** of these classical tune transformations transforms his song into Alessandro's:

- *Transposition*: increase or decrease every pitch in the sequence by the same amount;
- *Doubling (resp. halving)*: double (resp. halve) every duration in the sequence;
- *Retrogradation*: invert the sequence of notes, playing them from the last one to the first.

Note that each operation can be used at most once.

Help Giorgio file his claim on Alessandro's mischief as strongly as possible by finding the shortest sequence to change!

¹In the MINI (Musical Instrument Note Information) format.

📎 Among the attachments of this task you may find a template file `plagiarism.*` with a sample incomplete implementation.

Input

The first line contains the two integers N and V .

The next N lines contain two integers each (d_i and p_i) describing Giorgio's song.

The next N lines contain two integers each (d_i and p_i) describing Alessandro's song.

Output

You need to write a single line with one of the following contents:

- "SAME": if the two songs are identical;
- "TRANSFORMED L ": if Alessandro's song can be obtained by modifying a sequence of length L using some of the operations described above;
- "ORIGINAL L ": there is a sequence of length L that cannot be obtained just by applying the operations described above.

Constraints

- $1 \leq N \leq 100\,000$.
- $1 \leq V \leq 10^9$.
- $0 \leq d_i, p_i \leq V$ for each $i = 0 \dots N - 1$ for both songs.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

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|---|----------------------------|
| – Subtask 1 (0 points) | Examples. |
|  | |
| – Subtask 2 (20 points) | $N, V \leq 500$. |
|  | |
| – Subtask 3 (15 points) | $N \leq 500$. |
|  | |
| – Subtask 4 (25 points) | $N, V \leq 10\,000$. |
|  | |
| – Subtask 5 (10 points) | $V \leq 10\,000$. |
|  | |
| – Subtask 6 (30 points) | No additional limitations. |
|  | |

Examples

input	output
3 127 48 60 48 62 96 64 48 60 48 62 96 64	SAME
6 127 48 60 48 62 48 64 96 60 48 62 96 64 48 60 48 62 96 64 48 60 48 62 96 64	ORIGINAL 2
7 127 48 60 48 62 48 64 96 60 48 62 48 64 96 60 48 60 24 67 48 65 24 69 24 67 48 64 96 60	TRANSFORMED 4

Explanation

In the **first sample case**, the two songs are identical.

In the **second sample case**, Alessandro's song is obtained by rewriting the third and fourth notes 48 64, 96 60 into 96 64, 48 60, leaving the rest unchanged. Note that this is **not** obtainable through a combination of some classical tune transformations.

In the **third sample case**, Alessandro's song can be obtained by transforming the sequence from the second note to the fifth, by applying halving, retrogradation and then a +5 transposition.